CAPSTONE PROJECT

ONLINE QUIZ APPLICATION

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OUTLINE

- **Problem Statement** (Should not include solution)
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment (Step by Step Procedure)**
- **Result**
- **Conclusion**
- **Future Scope(Optional)**
- **References**

PROBLEM STATEMENT

Students and educators lack lightweight, interactive quiz tools for practice and testing. This project features a quiz app with multiple-choice questions, timers, and instant scoring. It supports learning through immediate feedback.

SYSTEM APPROACH

- Front-End Technologies:
  - HTML5: For structuring the content and layout of the quiz.
  - CSS3: For styling, enhancing the user interface, and making the application responsive.
  - JavaScript (Vanilla JS): For implementing interactive features such as timers, instant feedback, scoring, and navigation between questions.
- Advantages of the Chosen Approach:
  - Lightweight and browser-friendly (no server required).
  - Real-time interactivity with instant feedback.
  - Easy to extend and customize.

ALGORITHM & DEPLOYMENT

- Algorithm:
  - Initialize quiz data (questions, options, correct answers).
  - Display the first question with multiple-choice options.
  - Start a countdown timer for the quiz session.
  - Wait for the user to select an answer.
  - On selection, provide **instant feedback** (correct/wrong).
  - Enable the “Next” button only after an answer is selected.
  - Repeat steps 2–6 for all questions.
  - After the last question or when the timer ends:
    - Calculate the total score.
    - Display the result and feedback.
  - Allow the user to restart the quiz.

ALGORITHM & DEPLOYMENT

- **Deployment Procedure:**
- Create a folder structure:

```
OnlineQuiz/
    index.html
    style.css
    script.js
```

- Write the HTML, CSS, and JavaScript code as per the application design.
- Open index.html in any modern web browser (Chrome, Firefox, Edge).
- Test all functionalities: question display, timer, answer selection, feedback, scoring, and restart.
- (Optional) Host on a platform like **GitHub Pages** for online access.

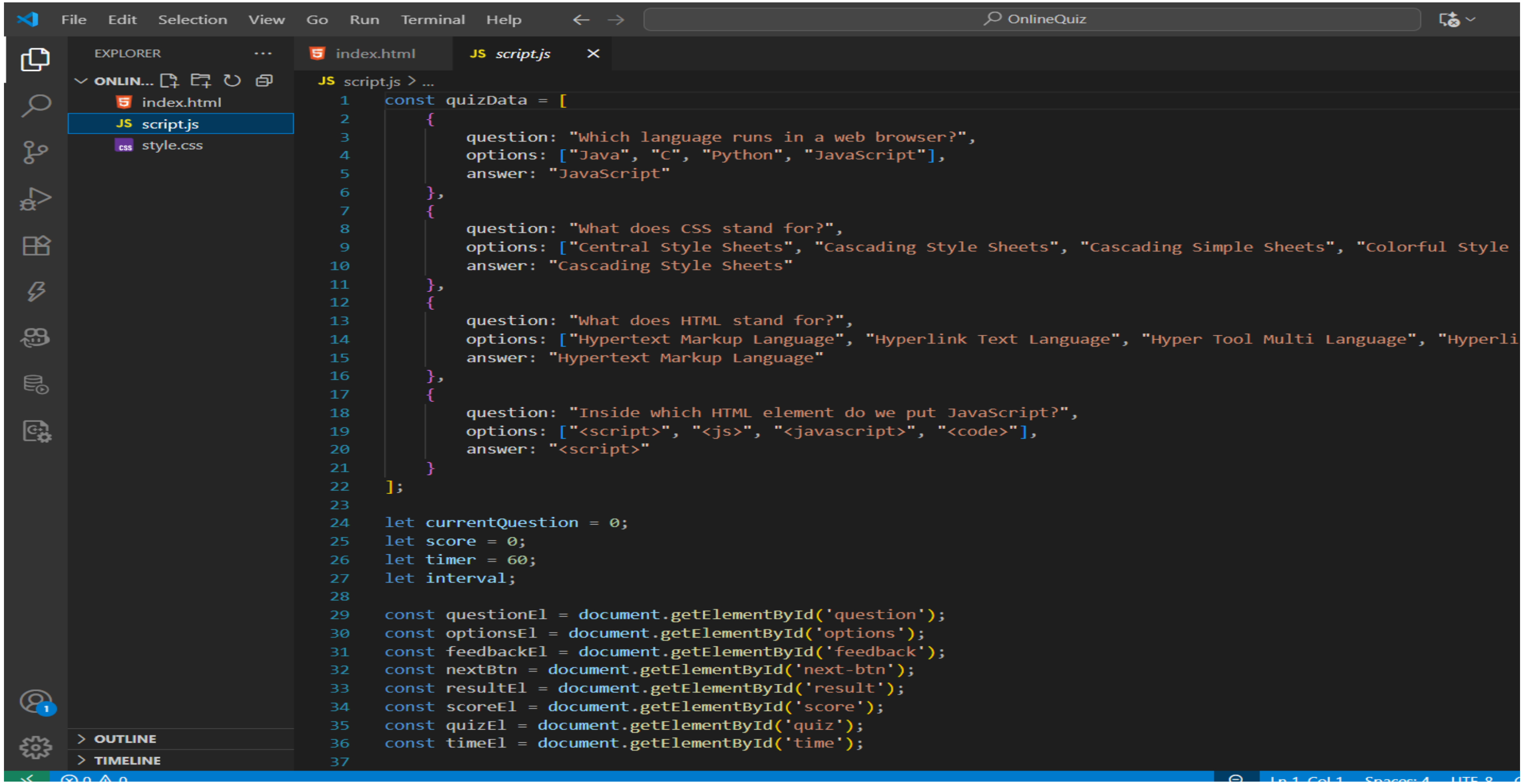
RESULT

```html
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Online Quiz App</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>
    <div class="quiz-container">
        <h1>Online Quiz Application</h1>
        <div id="quiz">
            <div id="question-container">
                <p id="question"></p>
            </div>
            <div id="options">

            </div>
            <div id="feedback"></div>
            <button id="next-btn">Next</button>
        </div>
        <div id="result" class="hide">
            <h2>Quiz Completed!</h2>
            <p id="score"></p>
            <button onclick="restartQuiz()">Restart Quiz</button>
        </div>
        <div id="timer">Time: <span id="time">60</span>s</div>
    </div>
    <script src="script.js"></script>
</body>

</html>
```

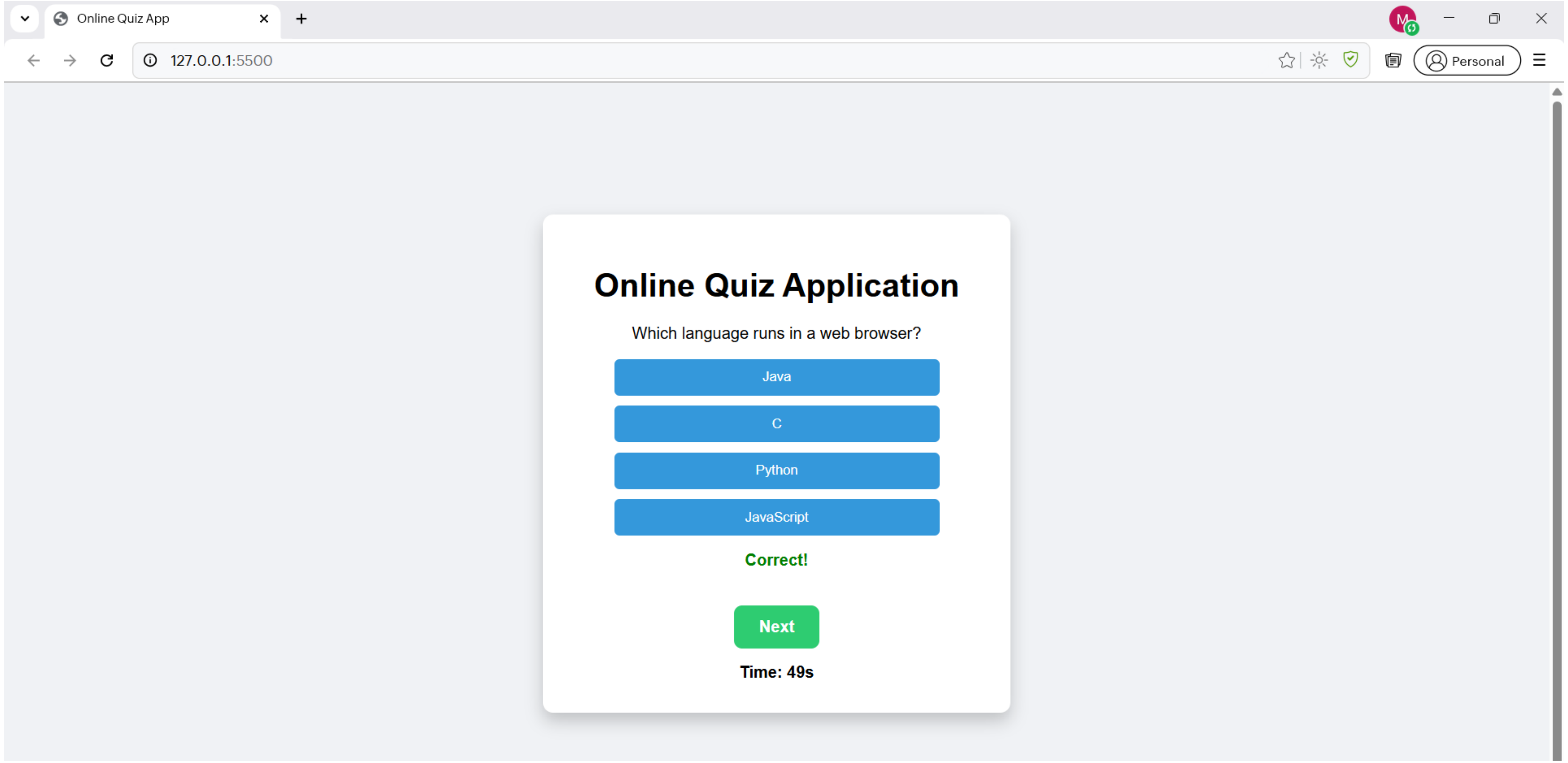
RESULT

```css
body {
    font-family: Arial, sans-serif;
    background-color: #f0f2f5;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
}

.quiz-container {
    background-color: white;
    padding: 30px;
    border-radius: 10px;
    box-shadow: 0 8px 16px rgba(0,0,0,0.2);
    width: 400px;
    text-align: center;
}

h1 {
    margin-bottom: 20px;
}

#options button {
    display: block;
    margin: 10px auto;
    padding: 10px 20px;
    width: 80%;
    border: none;
    border-radius: 5px;
    background-color: #3498db;
    color: white;
    cursor: pointer;
    transition: background-color 0.3s;
}

#options button:hover {
```

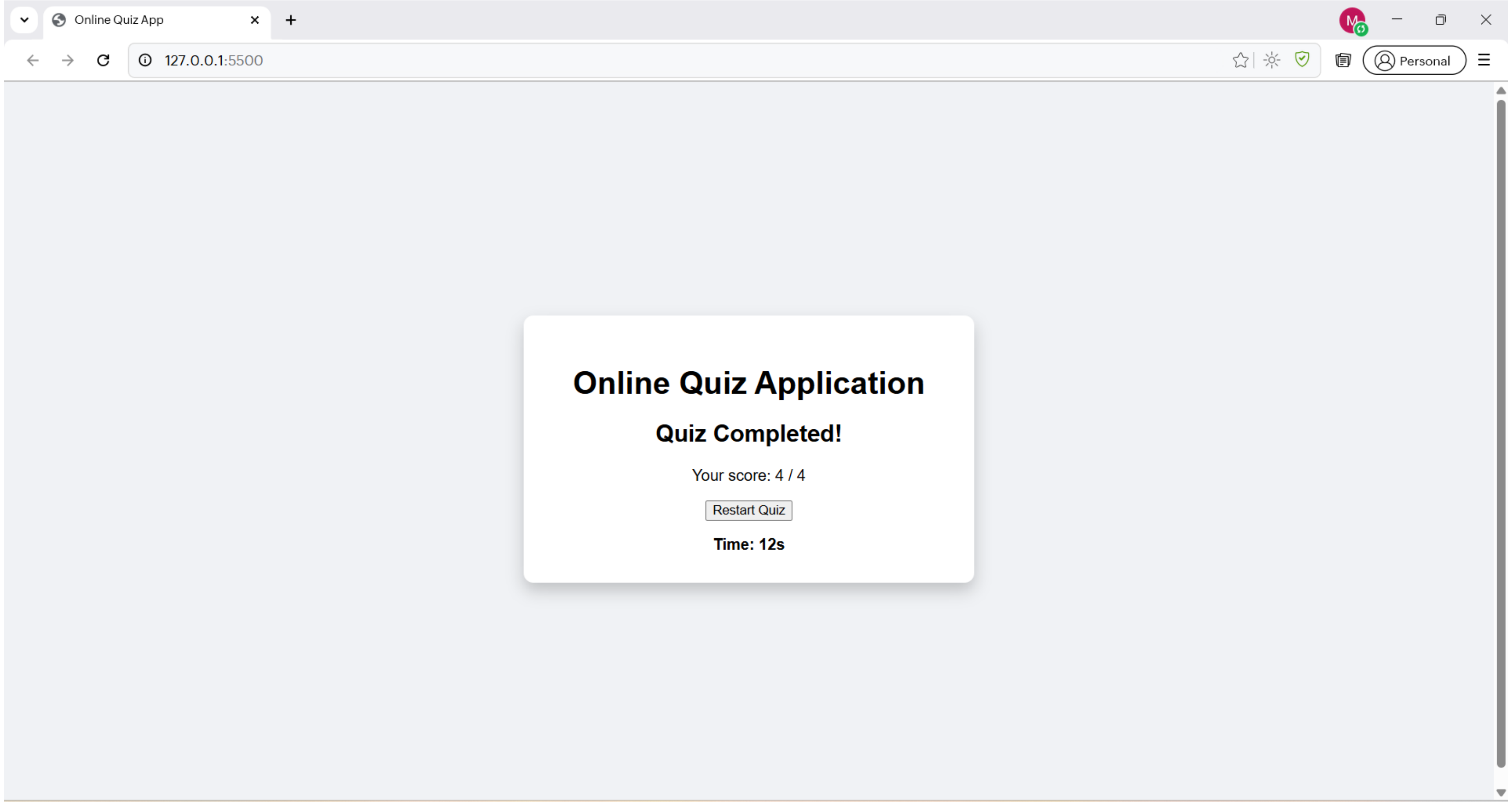
RESULT


File Edit Selection View Go Run Terminal Help
OnlineQuiz
EXPLORER
ONLIN...
index.html
script.js
style.css
OUTLINE
TIMELINE
index.html
script.js
script.js > ...
```
const quizData = [
    {
        question: "Which language runs in a web browser?",
        options: ["Java", "C", "Python", "JavaScript"],
        answer: "JavaScript"
    },
    {
        question: "What does CSS stand for?",
        options: ["Central Style Sheets", "Cascading Style Sheets", "Cascading Simple Sheets", "Colorful Style
        answer: "Cascading Style Sheets"
    },
    {
        question: "What does HTML stand for?",
        options: ["Hypertext Markup Language", "Hyperlink Text Language", "Hyper Tool Multi Language", "Hyperli
        answer: "Hypertext Markup Language"
    },
    {
        question: "Inside which HTML element do we put JavaScript?",
        options: ["<script>", "<js>", "<javascript>", "<code>"],
        answer: "<script>"
    }
];

let currentQuestion = 0;
let score = 0;
let timer = 60;
let interval;

const questionEl = document.getElementById('question');
const optionsEl = document.getElementById('options');
const feedbackEl = document.getElementById('feedback');
const nextBtn = document.getElementById('next-btn');
const resultEl = document.getElementById('result');
const scoreEl = document.getElementById('score');
const quizEl = document.getElementById('quiz');
const timeEl = document.getElementById('time');
```

RESULT


Online Quiz App
127.0.0.1:5500
Personal
Online Quiz Application
Which language runs in a web browser?
Java
C
Python
JavaScript
Correct!
Next
Time: 49s

RESULT



Online Quiz App
127.0.0.1:5500
Personal
Online Quiz Application
Quiz Completed!
Your score: 4 / 4
Restart Quiz
Time: 12s

GITHUB AND DEPLOYMNET LINK

- Github Link : https://github.com/mouli4401
- Deployment link: https://mouli4401.github.io/OnlineQuizFWD/

CONCLUSION

- The Online Quiz Application successfully provides an interactive and lightweight platform for learning and assessment. It allows students to practice multiple-choice questions with instant feedback and scoring. This approach enhances learning efficiency by giving immediate results and motivation to improve.

REFERENCES

- MDN Web Docs: HTML, CSS, JavaScript
- W3Schools: Quiz App Example
- Stack Overflow community discussions for interactive quiz implementation techniques

THANK YOU